HACKATHON ASSESSMENT-2

NAME : POOJA M

BUSSINESS SCENARIO-6

CREDIT CARD FRAUD DETECTION SYSTEM

SYSTEM DESCRIPTION

The bank monitors credit card transactions to detect potential fraud.

Transactions exceeding defined limits must be flagged.

DB2 TABLE STRUCTURE

CARD_TRANSACTION

Column	Type
TRANS_ID	INTEGER
CARD_NO	CHAR(16)
TRANS_AMOUNT	DECIMAL(9,2)
TRANS_DATE	DATE
STATUS	CHAR(1)

BUSINESS RULES

✓ Transaction amount > ₹1,00,000 → mark as **SUSPECT**

✓ Normal transactions marked **APPROVED**

✓ Suspicious transactions stored for compliance review

PROCESSING LOGIC

1. Receive transaction details
2. Insert transaction into DB2
3. Check amount threshold
4. Update status (APPROVED / SUSPECT)

EXPECTED OUTPUT

```
------ CARD TRANSACTION ------
Card Number : 4321 1111 2222 3333
Amount      : 150000
Status      : SUSPECT
Transaction Logged
------------------------------
```

COBOL +DB2

```
 IDENTIFICATION DIVISION.
 PROGRAM-ID. TRANSCA.
 DATA DIVISION.
 WORKING-STORAGE SECTION.
    EXEC SQL INCLUDE SQLCA END-EXEC.
    EXEC SQL INCLUDE CARDTRAN END-EXEC.
 01 WS-TRANS-ID     PIC S9(9).
 01 WS-CARD-NO      PIC X(16).
 01 WS-TRANS-AMOUNT PIC S9(7)V9(2).
 01 WS-TRANS-DATE   PIC X(10).
 01 WS-STATUS       PIC  X(10).
 01 WS-EOF          PIC  X(1).
 PROCEDURE DIVISION.
 000-MAIN-PROCESS.
      DISPLAY 'MAIN PARA'.
      MOVE 'N' TO WS-EOF.
      PERFORM 200-GET-PARA THRU 200-EXIT
      PERFORM 300-INSERT-PARA THRU 300-EXIT
      PERFORM 400-DISPLAY-PARA THRU 400-EXIT
      STOP RUN.
 200-GET-PARA.
     MOVE 1001 TO WS-TRANS-ID.
     MOVE WS-TRANS-ID TO DCL-TRANS-ID.
     MOVE '4321111122223333' TO WS-CARD-NO.
```

```
     MOVE WS-CARD-NO TO DCL-CARD-NO.
    MOVE 1500000 TO WS-TRANS-AMOUNT.
    MOVE WS-TRANS-AMOUNT TO DCL-TRANS-AMOUNT.
    MOVE '2026-03-10' TO WS-TRANS-DATE.
    MOVE WS-TRANS-DATE TO DCL-TRANS-DATE.
    IF WS-TRANS-AMOUNT > 100000
       MOVE 'SUSPECT' TO WS-STATUS
    ELSE
      MOVE 'APPROVED' TO WS-STATUS
    END-IF
    MOVE WS-STATUS TO DCL-STATUS.
200-EXIT.
   EXIT.
300-INSERT-PARA.
     DISPLAY 'INSERT PARA'
     EXEC SQL
       INSERT INTO Z82715.CARD_TRANSACTION
       (TRANS_ID,CARD_NO,TRANS_AMOUNT,TRANS_DATE,STATUS)
       VALUES(:DCL-TRANS-ID,:DCL-CARD-NO,:DCL-TRANS-AMOUNT,
              :DCL-TRANS-DATE,:DCL-STATUS)
     END-EXEC
     IF SQLCODE = 0
       DISPLAY "RECORD INSERTED SUCCESS"
     ELSE
       DISPLAY "INSERT FAILED,SQLCODE = " SQLCODE
       END-IF.
  300-EXIT.
     EXIT.
  400-DISPLAY-PARA.
      DISPLAY "-----CARD TRANSACTION----".
      DISPLAY "CARD NUMBER  : " DCL-CARD-NO.
```

DISPLAY "AMOUNT : " DCL-TRANS-AMOUNT.

DISPLAY "STATUS : " DCL-STATUS.

DISPLAY "TRANSACTION LOGGED".

400-EXIT.

EXIT.

DCLGEN GENERATED COBOL VARIABLE

```
*****************************************************************************
      EXEC SQL DECLARE Z82715.CARD_TRANSACTION TABLE
      ( TRANS_ID                          INTEGER,
        CARD_NO                           CHAR(16),
        TRANS_AMOUNT                      DECIMAL(9, 2),
        TRANS_DATE                        DATE,
        STATUS                            CHAR(10)
      ) END-EXEC.
*****************************************************************************
* COBOL DECLARATION FOR TABLE Z82715.CARD_TRANSACTION
*****************************************************************************
 01   DCL-CARDTRAN.
*                                 TRANS_ID
      10 DCL-TRANS-ID                  PIC S9(9) USAGE COMP-5.
```

```
*                                 CARD_NO
      10 DCL-CARD-NO                   PIC X(16).
*                                 TRANS_AMOUNT
      10 DCL-TRANS-AMOUNT              PIC S9(7)V9(2) USAGE COMP-3.
*                                 TRANS_DATE
      10 DCL-TRANS-DATE                PIC X(10).
*                                 STATUS
      10 DCL-STATUS                    PIC X(10).
*****************************************************************************
* THE NUMBER OF COLUMNS DESCRIBED BY THIS DECLARATION IS 5                  *
*****************************************************************************
************************** Bottom of Data ******************************
```

PRECOMPILE+COMPILE+BIND

```
//COSQM1     JOB (145),'CSQ COMP',CLASS=A,MSGCLASS=A,
//           MSGLEVEL=(1,1),TIME=(0,1)
//*******************INPUT AREA*************************
//  SET MEMNAME=TRANSCA
//  SET LOADMEM=TRANSCA
//  SET SOURCE=Z82715.POOJA.COBOL
//  SET INCLUDE=Z82715.POOJA.DCLGEN
//  SET DBRMLIB=Z82715.POOJA.DBRMLIB
//  SET LOADLIB=Z82715.POOJA.LOADLIB
//  SET COBCOPY=Z82715.POOJA.COBOL
//**************PRECOMPILE THE COBOL**************
//COSQMP1    EXEC PGM=DSNHPC,
//           PARM='HOST(IBMCOB),XREF,SOURCE,FLAG(I),APOST,NEWFUN(NO)'
//STEPLIB    DD DSN=DSND10.SDSNLOAD,DISP=SHR
//DBRMLIB    DD DSN=&DBRMLIB(&LOADMEM),DISP=SHR
//SYSIN      DD DSN=&SOURCE(&MEMNAME),DISP=SHR
//SYSCIN    DD DSN=&&DSNHOUT,DISP=(NEW,PASS,DELETE),UNIT=SYSDA,
//              SPACE=(800,(500,500))
//SYSLIB     DD DSN=&INCLUDE,DISP=SHR
//           DD DSN=&COBCOPY,DISP=SHR
//SYSPRINT   DD SYSOUT=*
//SYSTERM    DD SYSOUT=*
//SYSDUMP    DD SYSOUT=*
//SYSUT1     DD SPACE=(800,(500,500),,,ROUND),UNIT=SYSDA
```

```
//SYSUT2     DD  SPACE=(800,(500,500),,,ROUND),UNIT=SYSDA
//***************COBOL  COMPILE*********
//COBSTEP    EXEC  PROC=IGYWCL
//SYSIN      DD  DSN=&&DSNHOUT,DISP=(OLD,DELETE)
//LKED.SYSLIB  DD  DISP=SHR,DSN=DSND10.SDSNLOAD
//                 DD  DISP=SHR,DSN=CEE.SCEELKED
//                 DD  DISP=SHR,DSN=CEE.SCEERUN
//                 DD  DISP=SHR,DSN=ISP.SISPLOAD
//LKED.SYSLMOD  DD  DSN=&LOADLIB(&LOADMEM),DISP=SHR
```

```
//*************BIND  JOB*********************
//STEPBIND     EXEC  PGM=IKJEFT01,DYNAMNBR=64
//DBRMLIB      DD    DSN=&DBRMLIB,DISP=SHR
//STEPLIB      DD    DSN=&LOADLIB,DISP=SHR
//              DD    DSN=DSND10.SDSNLOAD,DISP=SHR
//SYSUT3       DD    UNIT=SYSDA,SPACE=(CYL,(500,50))
//SYSUT4       DD    UNIT=SYSDA,SPACE=(CYL,(500,50))
//SYSPRINT     DD    SYSOUT=*
//SYSTSPRT     DD    SYSOUT=*
//SYSTSIN      DD    *
     DSN  S(DBDG)
     BIND  PLAN(Z82715)  MEMBER(TRANSCA)  QUALIFIER(Z82715)-
     ACTION(REP)  RETAIN  ISOLATION(UR)  -
     VALIDATE(BIND)  ENCODING(EBCDIC)
     END
```

```
//CDBPGM     JOB  (125),'CURSOR RUN',CLASS=A,MSGCLASS=A,TIME=(0,1)
//CDBRUN     EXEC  PGM=IKJEFT01,DYNAMNBR=20
//STEPLIB    DD  DSN=CEE.SCEERUN,DISP=SHR
//           DD  DSN=DSND10.SDSNLOAD,DISP=SHR
//SYSTSPRT   DD  SYSOUT=*
//SYSPRINT   DD  SYSOUT=*
//SYSTSIN    DD  *
         DSN  SYSTEM(DBDG)
         RUN  PROGRAM(TRANSCA)  PLAN(Z82715)  LIB('Z82715.POOJA.LOADLIB')
         END
/*
```

OUTPUT

```
 SDSF  OUTPUT  DISPLAY  CDBPGM     JOB
 COMMAND  INPUT  ===>
*************************************
MAIN  PARA
INSERT  PARA
RECORD  INSERTED  SUCCESS
-----CARD  TRANSACTION----
CARD  NUMBER    :  4321111122223333
AMOUNT    :  150000000
STATUS    :  SUSPECT
TRANSACTION  LOGGED
*************************************
```